

Satpuda Core Table and Missing Column Audit Guide

Documentation-only schema review for new-user accounting, GST filing, supplier tracking, profit calculation, account clearing, and future sales/purchase returns. No source code changes included.

Generated April 25, 2026 from the current SQLite schema and expected future accounting needs.

1. Most Important Finding

Yes, supplier information should be linked to medicine stock. But the best design depends on how the medicine table is used.

Option	Recommendation	Why
Best normalized design	Create medicine_master and medicine_batches. Put supplier_id and purchase_item_id on medicine_batches.	A medicine can have many batches from different suppliers. Supplier belongs to the batch/source stock, not only the medicine name.
If keeping current medicines table	Add supplier_id, purchase_id, purchase_item_id, last_purchase_date, and cost_rate_ex_gst to medicines.	Current medicines rows already store batch_no, expiry_date, stock_qty, rate, and mrp, so they behave like batch/stock rows. Supplier should be stored on the same row.
Do not do only this	Do not add supplier TEXT only.	A plain supplier name can break when supplier spelling changes. Use supplier_id foreign key and optionally supplier_name_snapshot for old invoice display.

2. Global Column Rules

- Use foreign keys for relationships: supplier_id, customer_id, medicine_id, purchase_item_id, sale_item_id.
- Use status columns instead of deleting posted records: ACTIVE, VOID, RETURNED, PARTIAL_RETURN, CLOSED.
- For money, prefer integer paise columns or Decimal-safe NUMERIC/TEXT storage. SQLite REAL can create paise-level rounding errors.
- For every posted business table, add created_at, updated_at, created_by, updated_by, voided_at, void_reason where useful.
- For GST, store taxable_value, gst_rate, cgst_amount, sgst_amount, igst_amount, and line_total per item. Do not rely only on GST percent.
- For history clarity, use ledger tables for balances instead of rewriting account_cleared flags across old invoices.

3. Current Tables Found

Current app tables inspected: boxes, customers, doctors, medicine_shelf, medicines, pharmacy_profile, purchase_items, purchases, racks, sales, sales_items, sections, settings, shelf_settings, shelves, suppliers, temp_log. Current views: bills_cleared, accounts_cleared, supplier_due_status. Current triggers: sales and purchases after insert/update triggers.

4. Table-by-Table Missing Columns

4.1 medicines

Current important columns: name, type, stock_qty, unit, gst_percent, mrp, rate, manufacturer, batch_no, expiry_date, hsn_code, schedule, location, content_drug.

Missing column	Priority	Why needed
supplier_id	P0	Links each medicine/batch stock row to the supplier. Needed for supplier-wise inventory, purchase traceability, purchase returns, batch recall, and reorder decisions.
purchase_id	P0	Connects stock row back to the purchase bill that created it. Useful for audit and correcting purchase edits/voids.
purchase_item_id	P0	Best link for exact source line. Required for purchase return and profit/cost traceability.
cost_rate_ex_gst	P0	Needed for correct profit. Current rate may change; sale profit needs historical cost snapshot.
rate_including_gst or cost_rate_including_gst	P1	Useful when supplier bill or UI uses inclusive cost. Avoid ambiguity.
unit_factor / pack_size	P0	Required for tablet/bolus conversion and consistent sale/purchase stock math.
barcode / sku	P2	Faster search and future scanner support.
reorder_level / min_stock_qty	P1	Low stock alerts and reorder reports.

Missing column	Priority	Why needed
is_active	P1	Hide discontinued medicines without deleting history.
updated_at	P1	Audit medicine/batch edits.
supplier_name_snapshot	P2	Optional display helper if supplier master name changes. Keep supplier_id as source of truth.

Best future split: medicine_master stores name/type/content/manufacturer/default GST/HSN; medicine_batches stores batch_no, expiry, stock, supplier_id, purchase_item_id, cost, mrp, location.

4.2 suppliers

Current important columns: name, address, phone, gstin, dl_numbers, created_at.

Missing column	Priority	Why needed
email	P2	Supplier communication and invoice sharing.
contact_person	P2	Useful when supplier business has multiple contacts.
state_code / state_name	P0	GST CGST/SGST vs IGST decision for purchase filing.
pan	P2	Supplier compliance record.
drug_license_expiry	P1	Medical/drug supplier compliance tracking.
payment_terms_days	P1	Supplier due aging and payable reminders.
opening_balance	P0	New-user setup needs existing supplier payable or advance balance.
is_active	P1	Hide inactive suppliers without deleting historical purchases.
updated_at	P1	Audit supplier profile changes.

4.3 customers

Current important columns: name, phone, address, document_name, total_due, total_credit, last_updated.

Missing column	Priority	Why needed
gstin	P1	Needed if any B2B/customer GST invoice is required.
email	P2	Digital bill sharing.
state_code / state_name	P1	Needed for CGST/SGST vs IGST if selling outside state or to GST customers.
opening_balance	P0	New user can enter existing customer due/advance before first bill.
credit_limit	P2	Warning when customer due exceeds allowed limit.
is_active	P1	Hide old customers without deleting history.
updated_at	P1	Audit customer contact changes.

4.4 sales

Current important columns: bill_no, customer_id, bill_date, total_amount, discount, rounding, paid fields, due/credit fields, cleared flags, doctor_name.

Missing column	Priority	Why needed
invoice_status	P0	OPEN, PARTIAL, CLEARED, VOID, RETURNED, PARTIAL_RETURN. Cleaner than only bill_cleared/account_cleared.
payment_status	P0	UNPAID, PARTIAL, PAID, OVERPAID. Makes history filters simple.
taxable_total	P0	Required for GST filing and profit without GST.
cgst_total / sgst_total / igst_total / gst_total	P0	Header GST totals for quick reports and invoice audit.
item_discount_total	P0	Separates item discounts from bill discount.
bill_discount_type / bill_discount_input / bill_discount_amount	P0	Avoid current amount-vs-percent confusion on edit.
credit_used	P0	Tracks customer previous credit used against this sale.
return_total	P0	Needed when sales return is added.
balance_after_invoice	P1	Optional snapshot for history display; ledger remains source of truth.
voided_at / void_reason	P1	Safer than hard delete for posted bills.
created_by / updated_at	P1	Audit trail.

4.5 sales_items

Current important columns: sale_id, medicine_id, qty, rate, gst_percent, amount.

Missing column	Priority	Why needed
medicine_name_snapshot	P1	Old bill display should not change if medicine name changes.
batch_no_snapshot / expiry_date_snapshot	P0	Batch traceability, return validation, schedule reporting.
hsn_code	P0	GST reports need HSN at sale time.
unit_factor / pack_size	P0	Correct tablet/bolus quantity calculation.
gross_amount	P0	Base before discounts/tax extraction.
item_discount_type / input / amount	P0	Needed for edit, audit, and correct GST.
bill_discount_allocated	P0	GST filing requires bill discount allocated to item taxable values.
taxable_value	P0	Core GST report field.
cgst_amount / sgst_amount / igst_amount / gst_amount	P0	Output GST filing fields.
line_total	P0	Final item amount after discount and GST.
cost_rate_ex_gst_snapshot	P0	Correct profit calculation even if purchase cost changes later.
returned_qty / returned_amount	P0	Quick validation and history for sales return.

4.6 purchases

Current important columns: purchase_no, supplier_id, purchase_date, total_amount, bill_number, gst_calc_method, overall_discount, rounding, payment/due/credit/cleared fields.

Missing column	Priority	Why needed
supplier_bill_date	P1	Supplier bill date may differ from entry date. Important for GST period.
invoice_status / payment_status	P0	OPEN, PARTIAL, PAID, VOID, RETURNED states.
rate_tax_mode	P0	Marks purchase rates as GST-exclusive or GST-inclusive. Avoid formula ambiguity.
taxable_total	P0	Input GST filing base.
input_cgst_total / input_sgst_total / input_igst_total / input_gst_total	P0	Input GST report totals.
item_discount_total	P0	Separates item discounts from bill-level discount.
bill_discount_type / bill_discount_input / bill_discount_amount	P0	Avoid ambiguity of overall_discount amount vs percent.
credit_used	P0	Tracks supplier advance/credit applied to this purchase.
return_total	P0	Needed for purchase returns/debit notes.
balance_after_invoice	P1	Optional history snapshot. Ledger should remain source of truth.
voided_at / void_reason	P1	Safe correction path instead of hard delete.

4.7 purchase_items

Current important columns: purchase_id, medicine_id, qty, free_qty, type, hsn_code, gst_value, mrp, rate, manufacturer, batch_no, expiry_date, schedule, amount, discount_percent.

Missing column	Priority	Why needed
supplier_id	P1	Optional denormalized helper. Main supplier still comes from purchase header. Useful for faster supplier-wise item reports.
gross_taxable_amount	P0	Base before discount. Needed for audit.
discount_type / discount_input / discount_amount	P0	Supports both percent and rupee discounts without losing meaning.
bill_discount_allocated	P0	Correct GST after overall discount.
taxable_value	P0	Input GST filing field.
cgst_amount / sgst_amount / igst_amount / gst_amount	P0	Input GST report fields.

Missing column	Priority	Why needed
line_total	P0	Final line amount after discount and GST.
unit_factor / pack_size	P0	Stock and tablet/bolus conversion consistency.
stock_qty_added	P0	Stores actual stock units added including conversion and free qty.
returned_qty / returned_amount	P0	Purchase return validation and reports.
medicine_batch_id	P0	Directly links purchase item to created stock batch.

4.8 doctors

Current important columns: name, phone, registration_number, created_at.

Missing column	Priority	Why needed
clinic_name	P2	Doctor-wise reports and customer reference.
address	P2	Complete doctor record.
email	P2	Optional communication.
is_active	P1	Hide old doctors without deleting sales history.
updated_at	P1	Audit doctor profile changes.

4.9 pharmacy_profile

Current important columns: name, address, phone, email, gstin, dl_number, gst_enabled, logo_path.

Missing column	Priority	Why needed
state_code / state_name	P0	GST place-of-supply and CGST/SGST vs IGST decision.
pan	P2	Business compliance information.
invoice_prefix / purchase_prefix / return_prefix	P1	Controlled numbering for bills and returns.
financial_year_start_month	P1	GST/financial-year reports.
upi_id / bank_details	P2	Bill print/payment support.
signature_path	P2	Invoice print support.
updated_at	P1	Audit profile changes.

4.10 racks, sections, boxes, shelves, medicine_shelf

These tables support location/shelf management. Current columns are basic and enough for simple location names, but stronger inventory location support needs more structure.

Table	Missing column	Why needed
racks / sections / boxes / shelves	sort_order	Stable display order instead of name-only sorting.
racks / sections / boxes / shelves	is_active	Hide removed locations without breaking medicine history.
racks / sections / boxes / shelves	updated_at	Audit location changes.
medicine_shelf	qty_in_location	Needed if one batch can be split across locations.
medicine_shelf	is_primary	Default picking location.
medicine_shelf	assigned_at / updated_at	Audit location assignment changes.

4.11 settings and shelf_settings

Table	Missing column	Why needed
settings	module	Group settings by billing, GST, backup, UI, reports.
settings	data_type	Parse bool/int/decimal/date safely instead of guessing from text.
settings	description	Explain what each setting controls.
settings	updated_at	Audit setting changes.
shelf_settings	updated_at	Track when location display behavior changes.

4.12 temp_log

Current columns: message only. This looks like a temporary/debug table. For production, either remove it from the production schema or replace it with a real audit_log/app_log table.

Missing column	Priority	Why needed
level	P2	INFO/WARNING/ERROR classification.
source	P2	Which page/service created the log.
created_at	P1	A log without time is not useful.
context_json	P2	Structured debugging details.

5. Important Missing Tables

Some needs cannot be solved cleanly by adding columns to existing tables. These supporting tables are strongly recommended for new users.

Missing table	Priority	Why needed
party_ledger	P0	Single source of truth for customer/supplier due, credit, advance, and account clear.
payments	P0	Stores cash/online/refund/adjustment payments separately from invoices.
payment_allocations	P0	Shows which payment cleared which invoice. Required for clean history.
stock_ledger	P0	Every purchase/sale/return/void stock movement becomes auditable.
sale_returns and sale_return_items	P0	Future sales return with GST reversal and stock-in.
purchase_returns and purchase_return_items	P0	Future purchase return with input GST reversal and stock-out.
audit_log	P1	Track edits, voids, corrections, and who did them.
invoice_sequences	P1	Reliable bill numbers by financial year and document type.
medicine_master and medicine_batches	P1	Cleaner long-term replacement for one medicines table mixing master data and batch stock.

6. Views and Triggers

Current views/triggers are not missing columns, but they matter for schema design.

- Existing bill/account cleared triggers duplicate business logic that should move into a shared service or ledger posting layer.
- For new-user design, keep triggers only for simple safety validation or timestamp updates. Avoid calculating account due inside triggers and UI at the same time.
- Replace accounts_cleared/supplier_due_status style views with ledger-based views: current_customer_balances, current_supplier_balances, invoice_remaining_due, gst_sales_summary, gst_purchase_summary.

7. Supplier Column Recommendation Summary

If you want the simplest practical change later, add supplier_id to medicines. But for correct purchase return, batch traceability, and profit, add the full source link set.

Column	Add to	Why
supplier_id	medicines or medicine_batches	Main supplier/source for that batch stock.
purchase_id	medicines or medicine_batches	Which purchase bill created this stock.
purchase_item_id	medicines or medicine_batches	Exact purchase line for cost, GST, return, and audit.
cost_rate_ex_gst	medicines or medicine_batches	Profit and stock valuation.
stock_qty_available	medicine_batches if split design	Batch-level available stock.

8. Implementation Priority

Phase	Do this first
Phase 1	Add/design ledger tables, payments, stock_ledger, return tables, and item GST columns.
Phase 2	Add supplier source columns to medicine batch/stock rows: supplier_id, purchase_id, purchase_item_id.
Phase 3	Update Sales and Purchase save logic to populate all tax, discount, cost, ledger, and stock fields.

Phase	Do this first
Phase 4	Rebuild Sales History and Purchase History to read from snapshots and ledgers.
Phase 5	Build GST and profit reports from stored item facts, not current medicine values.

9. Final Recommendation

For new users, the database should be return-ready and GST-ready from the first bill. The most urgent missing medicine column is `supplier_id`, but the safer complete solution is `supplier_id + purchase_id + purchase_item_id` on the batch/stock row. Across sales and purchases, the most important missing columns are item-level taxable values, GST split amounts, discount input/amount fields, cost snapshots, status columns, and ledger/payment allocation tables. These will keep Sales History and Purchase History clear even after payments, credits, edits, voids, and future returns.